Nonstarch polysaccharide consumption in four Scandinavian populations.

Nonstarch polysaccharide (NSP) intake was measured in representative samples of 30 men aged 50-59 in 2 urban and 2 rural Scandinavian populations that exhibited a 3-4 fold difference in incidence of large bowel cancer. Intake was measured by chemical analysis of complete duplicate portions of all food eaten over one day by each individual. NSP intakes showed a rural-urban gradient, with 18.4 +/- 7.8 g/day in rural Finland and 18.0 +/- 6.4 g/day in rural Denmark versus 14.5 +/- 5.4 g/day in urban Finland and 13.2 +/- 4.8 g/day in urban Denmark. NSP intakes were also calculated (using food tables) from weighed food records kept over 4 days, one of which was the day on which the duplicate collection was made. Intakes were 2-2.5 g/day higher with this method than with direct chemical analysis, mainly because published tables of values have become outdated and inaccurate as a result of improved methods for measuring NSP in food. Individual variation from day to day in NSP intake was considerable. Average NSP intake and intake of some of its component sugars were inversely related to colon cancer incidence in this geographical comparison. To show a relationship at the individual level between diet and cancer risk in a prospective study would require detailed and accurate methods for the assessment of NSP consumption.